

SILAS M. FALDE

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SUMMARY

Machine learning engineering student at the University of Michigan with internship and research experience building Python NLP and LLM pipelines, data quality tooling, and data-driven analysis for real-world datasets. Comfortable with PyTorch, Scikit-learn, and modern Python workflows for model evaluation, preprocessing, and experimentation.

EDUCATION

University of Michigan School of Engineering

Junior B.E.S. in Data Science, GPA: 3.09/4.00

Ann Arbor, MI

August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

EXPERIENCE

Ally Financial, Inc.

Intern, Artificial Intelligence & Advanced Analytics

Detroit, MI

May 2026 – August 2026

- Prototyped large language model workflows to automate internal tasks and improve efficiency and safety.
- Built Python NLP pipelines to evaluate PII masking and redaction model performance for sensitive company documents.
- Enhanced a data dictionary migration tool by adding batch processing, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

University of Michigan Department of Epidemiology

Research Assistant – Computational Lead

Ann Arbor, MI

September 2025 – May 2026

- Led the technical development of a Python pipeline using a large language model to classify narratives of collective violence.
- Performed data cleaning, exploratory data analysis, natural language processing, and descriptive statistical analysis on a large sensitive dataset.
- Evaluated model performance with classification metrics to measure accuracy and identify opportunities for improvement.

Ally Financial, Inc.

Intern, Data Management

Detroit, MI

May 2025 – August 2025

- Created data dictionaries and lineage documentation in Python to improve data governance and compliance.
- Added data quality checks to existing data pipelines with Python and SQL to improve data integrity and reliability.
- Developed a Power BI data availability dashboard to monitor refresh rates and availability for critical data assets.

University of Michigan Department of Epidemiology

Research Assistant

Ann Arbor, MI

September 2024 – May 2025

- Earned 3rd place in the CDC Youth Mental Health: Novel Variables Data Challenge, placing in roughly the top 1% of entries.
- Developed Python pipelines using a large language model on a high-performance computing cluster to extract trends from more than 400,000 suicide victim narratives.
- Cleaned, preprocessed, and engineered features from a confidential dataset of more than 400,000 narratives.
- Documented findings with reports and visualizations for the research team and publication.

PROJECTS

FPS PvP Dynamics

Project Developer

Ann Arbor, MI

March 2026 – April 2026

- Built a configurable **agent-based simulation** in Python to model player behavior, combat, and objectives on a grid with stochastic state updates, rule-based decisions, and per-tick match tracking.
- Developed data collection and analysis tooling with **trace capture**, **CSV export**, and **visualization** for debugging and validation.
- Ran **parameter sweeps**, replication studies, **Latin hypercube sampling**, and **Sobol sensitivity analysis** to evaluate behavior and key parameters.

Photo Processing Helper

Project Developer

Ann Arbor, MI

Jan 2026 - Present

- Built a Python image-processing toolkit to automate photo framing, collage generation, panorama stitching, and RAW-to-JPG matching across large image folders.
- Implemented computer vision pipelines with OpenCV, NumPy, Pillow, and rawpy to resize, crop, align, and blend images while preserving exact output dimensions and high-bit-depth detail; added validation and testing around file discovery, output integrity, and stitching behavior to keep the pipelines reliable for real-world photo datasets.
- Designed a modular codebase with shared library logic, CLI tools, and a notebook workflow to support repeatable batch processing and interactive experimentation.

SKILLS

Programming: Python, C++, R, SQL, Java

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, spaCy, NLTK

Software: Git, VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta